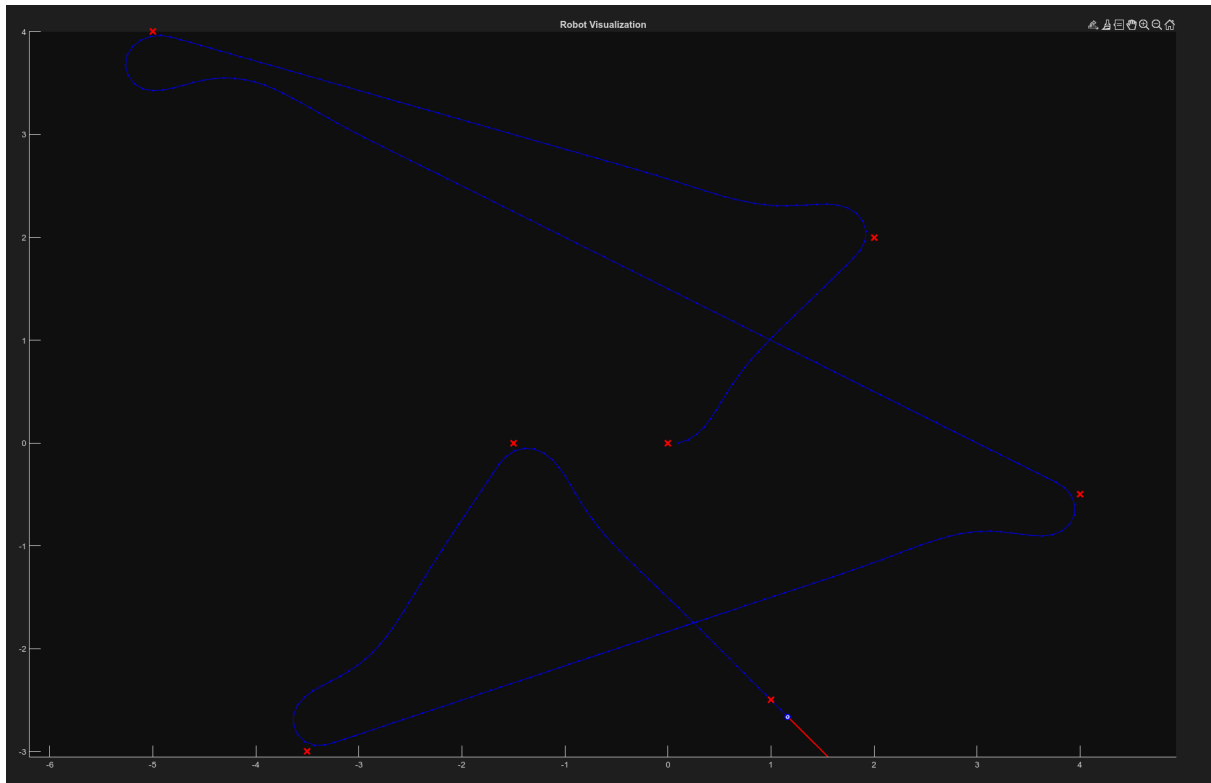


Figura 1:

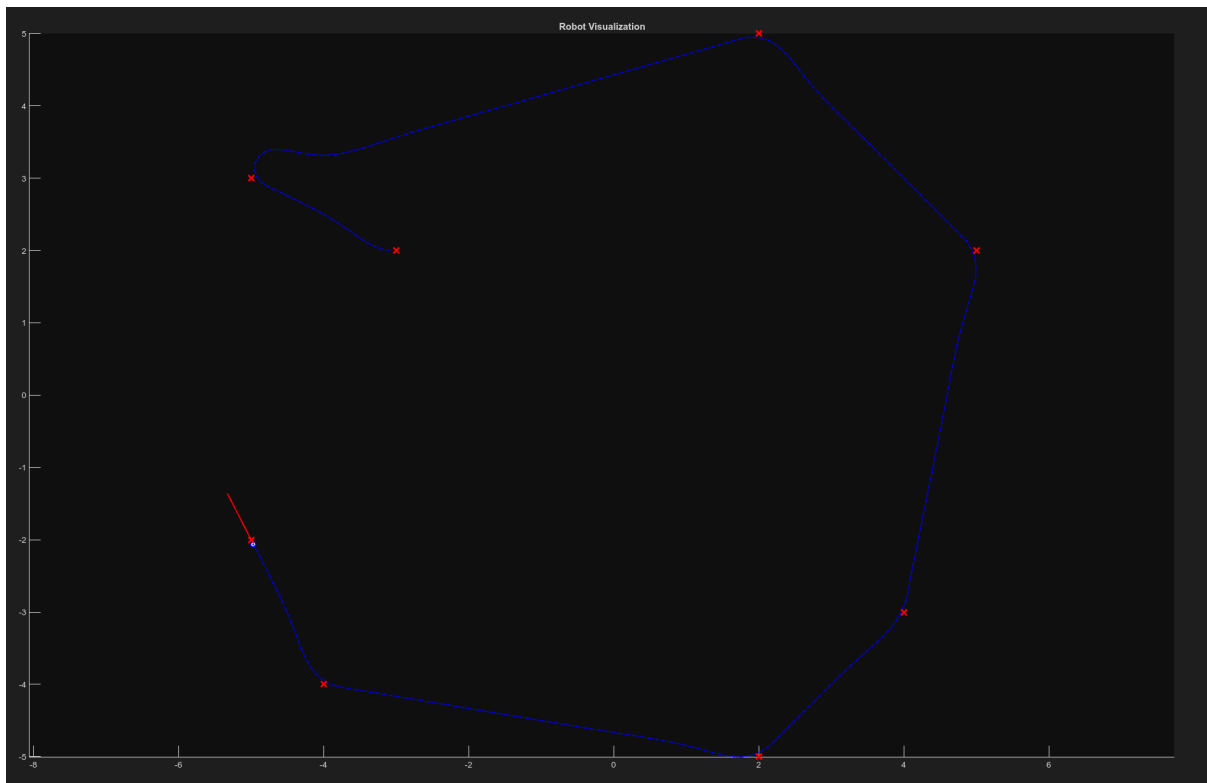


```
sampleTime = 0.1;           % Sample time [s]
tVec = 0:sampleTime:36.5;   % Time array
initPose = [0;0;0];         % Initial pose (x y theta)

% Define waypoints
waypoints = [0,0; 2,2; -5,4; 4,-0.5; -3.5,-3; -1.5,0; 1,-2.5];

%% Pure Pursuit Controller
controller = controllerPurePursuit;
controller.Waypoints = waypoints;
controller.LookaheadDistance = 0.45;
controller.DesiredLinearVelocity = 1;
controller.MaxAngularVelocity = 4;
```

Figura 2:



%% Simulation parameters

```
sampleTime = 0.1;           % Sample time [s]
tVec = 0:sampleTime:30;     % Time array
initPose = [-3;2;pi];       % Initial pose (x y theta)
```

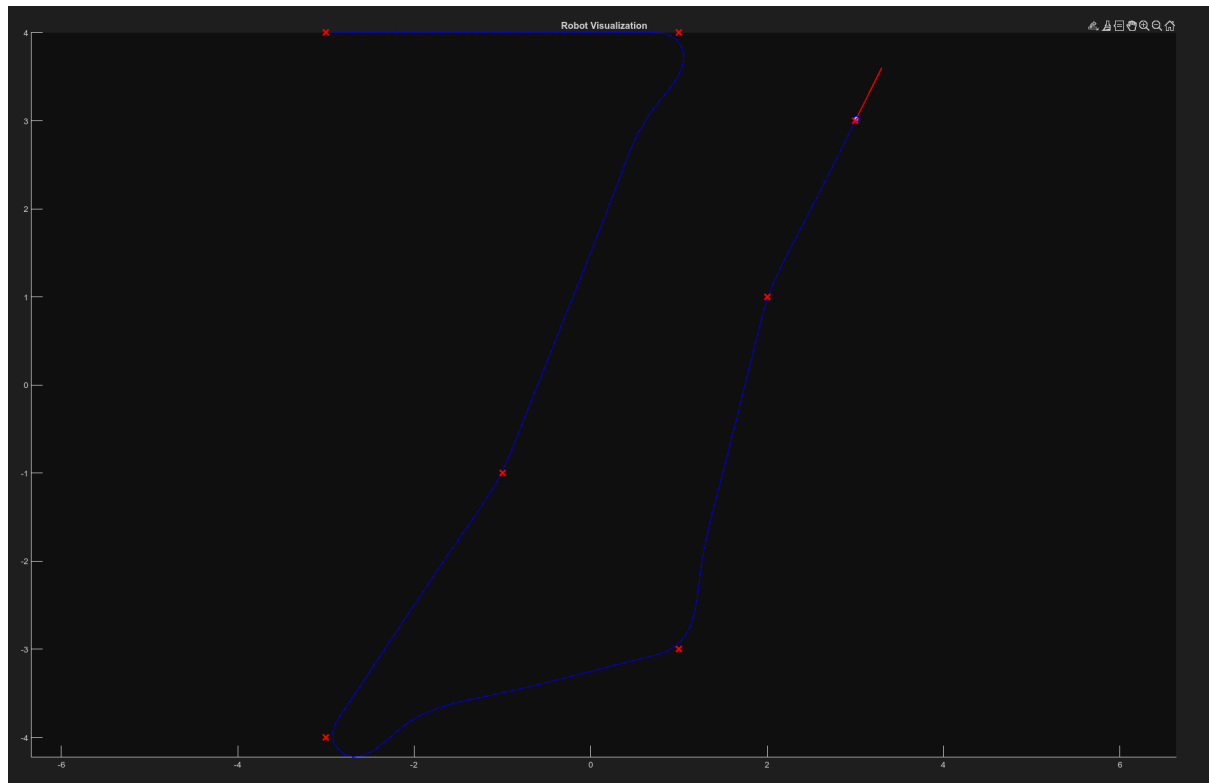
% Define waypoints

```
waypoints = [-3,2; -5,3; 2,5; 5,2; 4,-3; 2,-5; -4,-4; -5,-2];
```

%% Pure Pursuit Controller

```
controller = controllerPurePursuit;
controller.Waypoints = waypoints;
controller.LookaheadDistance = 0.45;
controller.DesiredLinearVelocity = 1;
controller.MaxAngularVelocity = 4;
```

Figura 3:



%% Simulation parameters

```
sampleTime = 0.1;           % Sample time [s]
tVec = 0:sampleTime:24;     % Time array
initPose = [-3;4;0];        % Initial pose (x y theta)
```

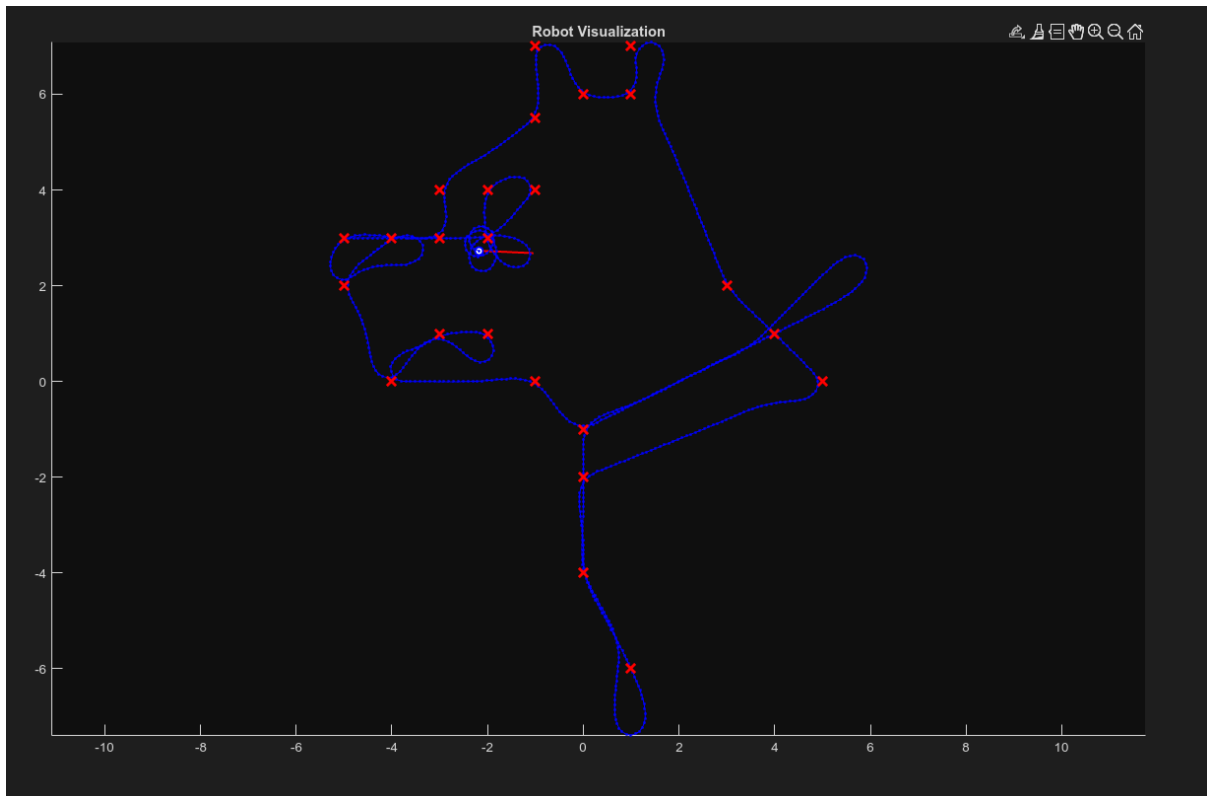
% Define waypoints

```
waypoints = [-3,4; 1,4; -1,-1; -3,-4; 1,-3; 2,1; 3,3];
```

%% Pure Pursuit Controller

```
controller = controllerPurePursuit;
controller.Waypoints = waypoints;
controller.LookaheadDistance = 0.45;
controller.DesiredLinearVelocity = 1;
controller.MaxAngularVelocity = 4;
```

Figura 4:



%% Simulation parameters

```
sampleTime = 0.1;           % Sample time [s]
tVec = 0:sampleTime:82;     % Time array
initPose = [-5;3;0];        % Initial pose (x y theta)
```

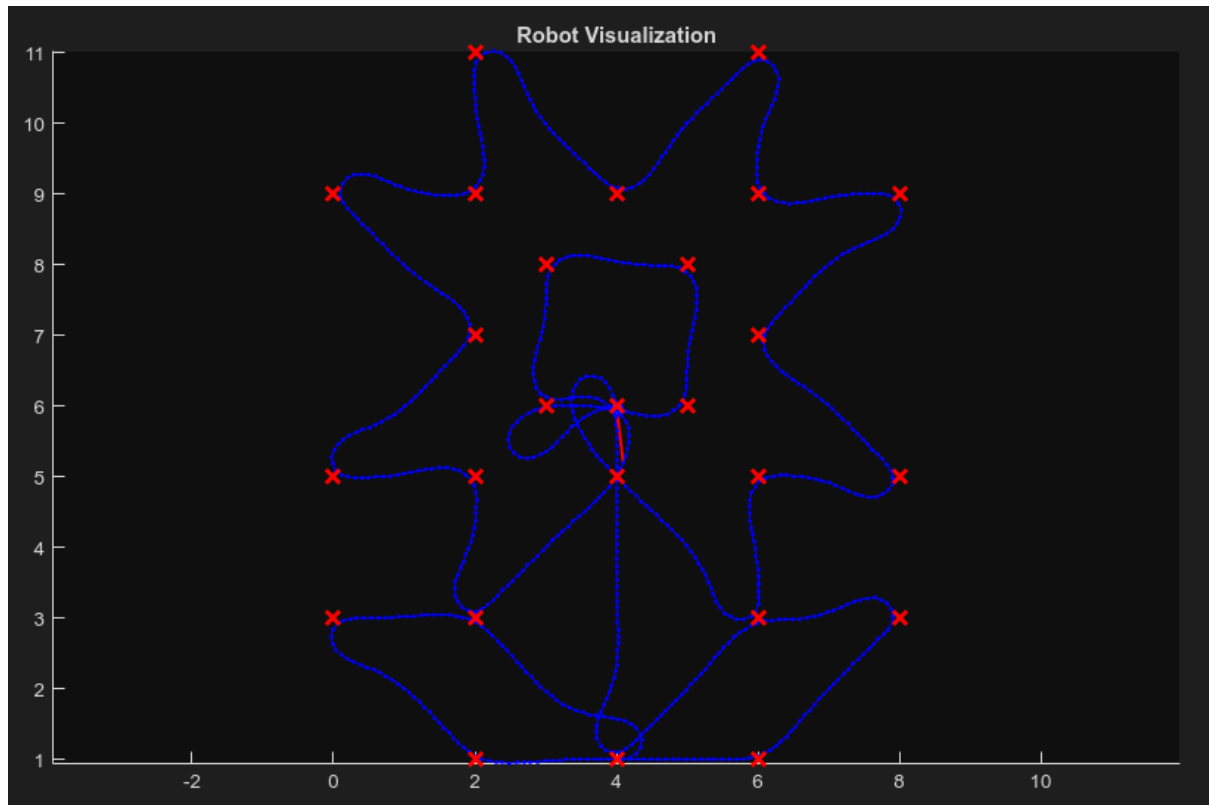
% Define waypoints

```
waypoints = [-5,3; -3,3; -3,4; -1,5.5; -1,7; 0,6; 1,6; 1,7; 3,2; 5,0; 0,-2; 0,-4; 1,-6; 0,-4; 0,-1; 4,1; 0,-1; -1,0; -4,0; -3,1; -2,1; -3,1; -4,0; -5,2; -4,3; -5,2; -5,3; -2,3; -1,4; -2,4; -2,3];
```

%% Pure Pursuit Controller

```
controller = controllerPurePursuit;
controller.Waypoints = waypoints;
controller.LookaheadDistance = 0.5;
controller.DesiredLinearVelocity = 1;
controller.MaxAngularVelocity = 4;
```

Figura 5:



%% Simulation parameters

```
sampleTime = 0.1;           % Sample time [s]
tVec = 0:sampleTime:79;     % Time array
initPose = [4;1;0];         % Initial pose (x y theta)
```

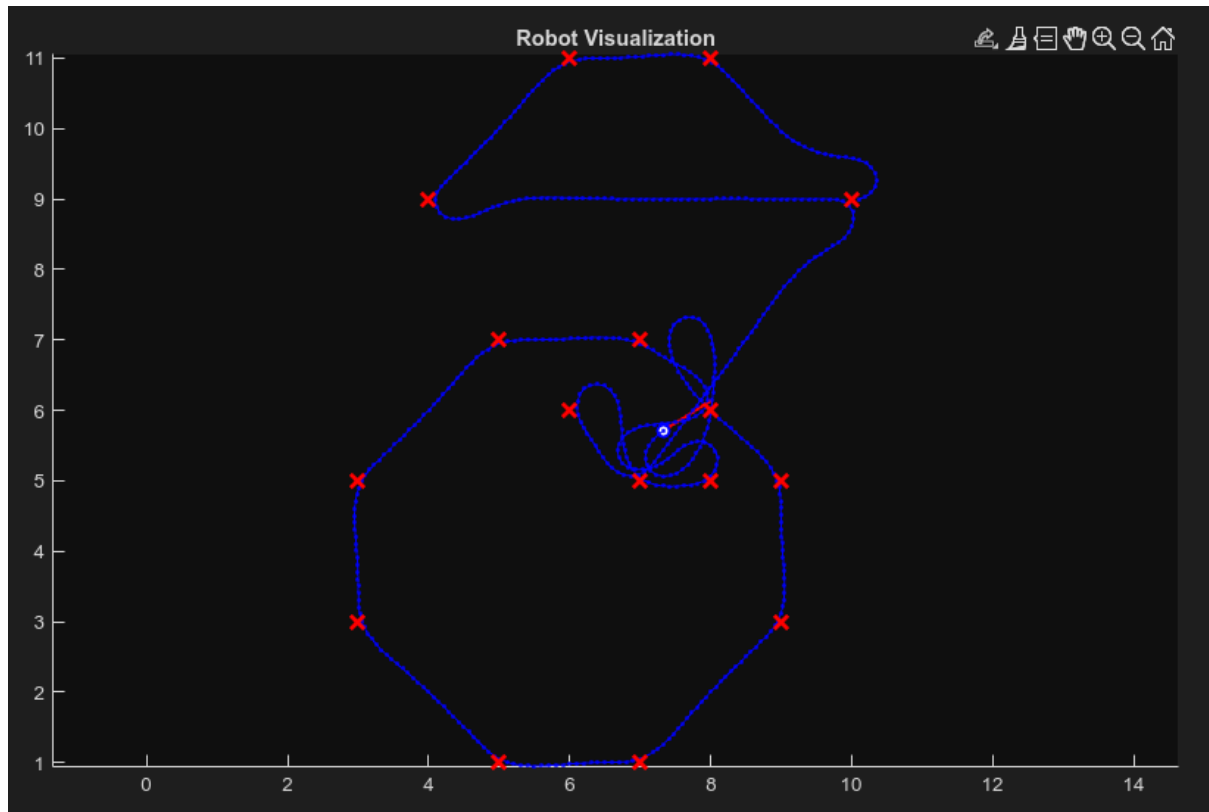
% Define waypoints

```
waypoints = [4,1; 2,3; 0,3; 2,1; 6,1; 8,3; 6,3; 4,1; 4,6; 3,6; 3,8; 5,8; 5,6; 3,6; 4,6; 4,5; 2,3; 2,5;
0,5; 2,7; 0,9; 2,9; 2,11; 4,9; 6,11; 6,9; 8,9; 6,7; 8,5; 6,5; 6,3; 4,5];
```

%% Pure Pursuit Controller

```
controller = controllerPurePursuit;
controller.Waypoints = waypoints;
controller.LookaheadDistance = 0.5;
controller.DesiredLinearVelocity = 1;
controller.MaxAngularVelocity = 4;
```

Figura 6:



%% Simulation parameters

```
sampleTime = 0.1;           % Sample time [s]
tVec = 0:sampleTime:52;     % Time array
initPose = [10;9;0];        % Initial pose (x y theta)
```

% Define waypoints

```
waypoints = [10,9; 8,11; 6,11; 4,9; 10,9; 7,5; 6,6; 7,5; 8,5; 7,5; 8,6; 7,7; 5,7; 3,5; 3,3; 5,1; 7,1;
9,3; 9,5; 8,6];
```

%% Pure Pursuit Controller

```
controller = controllerPurePursuit;
controller.Waypoints = waypoints;
controller.LookaheadDistance = 0.5;
controller.DesiredLinearVelocity = 1;
controller.MaxAngularVelocity = 5;
```